

Business Requirements Document

CULTURESHIFT AI — MULTI-AXIS AUDIO STORY ADAPTATION ENGINE

Project Name:	CultureShift AI (PocketFM Hackathon)	Version:	1.0
Author:	Product Owner & Business Analyst Team	Date:	July 2026
Target Audience:	Engineering Team, AI/Prompt Engineers, Hackathon Judges		

1. EXECUTIVE SUMMARY & BUSINESS OBJECTIVES

1.1 Project Overview

CultureShift AI is an intelligent story adaptation and hyper-localization engine designed specifically for audio streaming platforms like PocketFM. By decoupling a narrative's underlying structural skeleton from its surface presentation, the platform enables creators and production teams to automatically transform a single story IP across both **Genre Target** and **Cultural/Regional Flavour**. The system maintains strict plot integrity while generating hyper-local dialogue, regional idioms, custom voice accents, and high-suspense promotional trailers.

1.2 Business Objectives

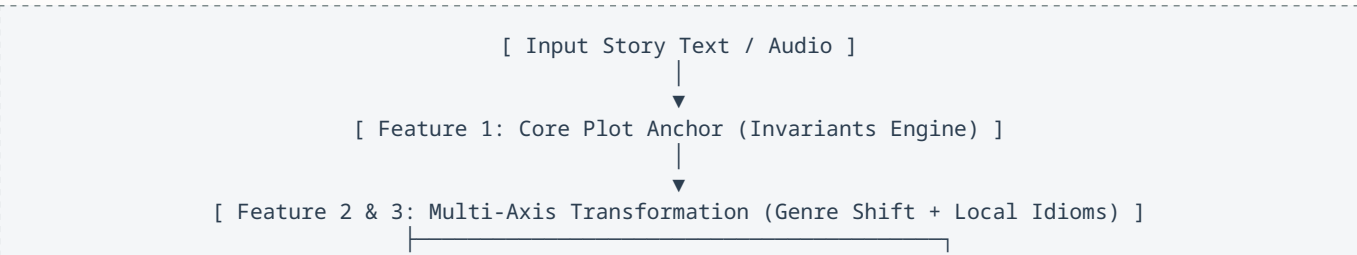
- **Scalable Localization:** Reduce the time and cost required to adapt successful audio stories across regional demographics by up to 90%.
- **Increased Listener Retention:** Maximize episode completion rates by instantly matching story tropes, humor, idioms, and accents directly to regional listener preferences.
- **Rapid Prototyping & Distribution:** Enable creators and content executives to test multiple genre and region variations of a story prior to full-scale production commitment.

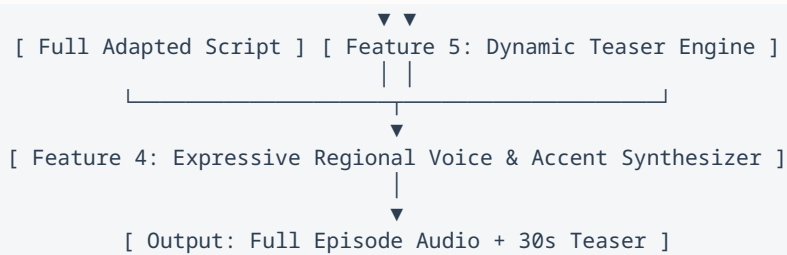
2. TARGET THEMES & PROBLEM STATEMENT ALIGNMENT

PocketFM Theme	Strategic Alignment & Implementation
P5: Creator Superpowers AI Rewrite Engine	Transforms base stories across genres (Horror, Comedy, Thriller, Romance) while locking down core plot beats. Layers hyper-local cultural context, slang, and suspense hooks on top of the genre transformation.
P6: AI Agents Entertainment CEO Agent	Functions as an autonomous decision-making and localization agent that analyzes IP performance metrics and automatically scales a single story across hundreds of micro-regions and dialects.

3. SYSTEM ARCHITECTURE & WORKFLOWS

The system relies on an isolated pipeline where structural analysis is decoupled from stylistic rewriting and audio synthesis:





4. FUNCTIONAL REQUIREMENTS & SPECIFICATIONS

Feature 1: Core Plot Anchor (Invariants Engine)

Description: Extracts and locks the foundational structural skeleton of the story before any rewrite takes place.

Functional Rules: Must extract and lock five core invariants: *Inciting Incident*, *Key Plot Beats*, *Character Motivations*, *Climax*, and *Narrative Resolution*. Outputs a structured JSON object.

Acceptance Criteria: Transformed output script must pass an automated verification check confirming all five core plot invariants remain 100% consistent with the original IP.

Feature 2: Multi-Axis Transformation Matrix

Description: A dual-axis processing layer allowing simultaneous selection of target Genre and target Regional Culture.

Supported Target Axes:

- **Genre Axis:** Horror, Comedy, Thriller, Romance, Sci-Fi, Drama.
- **Cultural/Regional Axis:** Rural Bhojpuri, Texas Country, South London Grime, Street Lagos Pidgin, Mumbai Tapri, etc.

Acceptance Criteria: Output script demonstrates atmospheric, pacing, and stylistic tropes of both selected parameters without narrative drift.

Feature 3: Hyper-Local Idiom & Humor Mapper

Description: Semantic rewrite layer that translates generic concepts, idioms, and settings into hyper-local equivalents.

Functional Rules: Automatically identifies and replaces non-translatable metaphors, local jokes, regional food/drink items, and geographical references (e.g., *"Grabbing a Starbucks in Manhattan"* → *"Drinking a Cutting Chai at a Mumbai Tapri"* or *"Grabbing a Sweet Tea in Georgia"*).

Acceptance Criteria: Zero culturally out-of-place idioms remain in the final script while original emotional weight is preserved.

Feature 4: Expressive Regional Voice & Accent Synthesizer

Description: Text-to-Audio conversion engine tuned for regional speech patterns, cadences, and local accents.

Acceptance Criteria: Generated audio sounds like an authentic native speaker from the target region, complete with local vocal inflections, rather than a generic TTS translation voice.

Feature 5: Dynamic Suspense Teaser & Hook Generator

Description: An automated trailer creation engine that distills the converted story into a high-impact, 30–45 second audio trailer.

Functional Rules: Identifies the highest-tension plot point/cliffhanger and structures a hook: *Intro Hook* → *Rising Tension* → *Cliffhanger Question* in the target genre, culture, and accent.

Acceptance Criteria: Outputs a complete, standalone 30–45 second high-suspense audio trailer.

5. NON-FUNCTIONAL & TECHNICAL GUIDELINES

- **Latency & Speed:** Teaser script generation must complete in under 10 seconds to ensure seamless live demo execution.
- **Modularity:** System components (Plot Extractor, Script Transformer, TTS Engine) must operate as independent modules via structured JSON APIs.
- **Development Guidelines:**
 - **LLM Layer:** Use strict JSON schemas (e.g., instructor) for Feature 1 (Plot Anchor).
 - **Prompting Layer:** Implement system prompts with specific target persona guidelines for Feature 2 & 3.
 - **TTS Layer:** Leverage accent-capable TTS providers (e.g., ElevenLabs, Azure Neural TTS) with specific voice IDs.

6. SUCCESS METRICS & HACKATHON DEMO CRITERIA

Metric / Criteria	Target Standard
Narrative Consistency	100% preservation of core plot resolution and character motivations across transformations.
Cultural Authenticity	Accurate adaptation of slang, humor, and regional references without direct literal translation artifacts.
Audio & Demo Impact	High-quality accented voice performance with a compelling 30-second suspense trailer for live presentation.